Arabic Spoken Command Recognition Using MFCC and Classical Machine Learning Techniques

Noor Mohammad [1], *Dana Abu Rayya*[2] ,*Taima Nazzal* [3]

[1] Department of Electrical and Computer Engineering, Birzeit University, Palestine
1222703@student.birzeit.edu,1210195@student.birzeit.edu,1220701@student.birzeit.edu

Abstract

Arabic speech recognition remains a challenging task due to dialectal variations, pronunciation differences, and the limited availability of labeled datasets. This project presents an isolated Arabic spoken command recognition system based on classical speech processing and machine learning techniques. A small dataset consisting of six Arabic commands (ابدأ، افتح، اغلق، يسار ، يمين، توقف) was collected from multiple speakers in a quiet environment and recorded at a sampling rate of 16 kHz. Each audio signal was preprocessed through silence removal and amplitude normalization. Mel-Frequency Cepstral Coefficients (MFCCs) were extracted to represent the acoustic characteristics of speech signals, and statistical features including mean and standard deviation were used to form fixed-length feature vectors. Several machine learning classifiers were evaluated, including K-Nearest Neighbors (KNN), Gaussian Mixture Models (GMM), Random Forest, and Support Vector Machine (SVM). Experimental results demonstrate that the SVM classifier achieved the highest recognition accuracy, reaching approximately 95.56%, outperforming the other models. Confusion matrix analysis shows that most commands are recognized correctly, with minor confusion between phonetically similar words. The results confirm the effectiveness of MFCC features combined with classical machine learning methods for isolated Arabic spoken command recognition.

Index Terms— Arabic speech recognition, MFCC, spoken commands, machine learning, SVM

1. Introduction

Speech-based human–computer interaction has become an essential component of modern intelligent systems, enabling users to interact with devices in a natural and efficient manner. Spoken command recognition, in particular, is widely used in applications such as smart home control, assistive technologies, and voice-operated systems. While speech recognition systems have achieved high performance for widely spoken languages, recognizing Arabic speech remains a challenging task.

Arabic speech recognition faces several difficulties, including dialectal variation, pronunciation diversity, and the limited availability of publicly accessible labeled datasets. These challenges are more evident in small-scale tasks such as isolated spoken command recognition, where system performance highly depends on feature representation and classifier selection. Therefore, developing efficient and reliable recognition systems using limited data is an important research problem.

In this project, we focus on recognizing isolated Arabic spoken commands using classical speech processing techniques and machine learning classifiers. A small dataset consisting of six commonly used Arabic commands is collected from multiple speakers in a controlled environment. The speech signals are first preprocessed to remove silence and normalize amplitude. Acoustic features are then extracted using Mel-Frequency Cepstral Coefficients (MFCC), which are widely known for their effectiveness in representing speech characteristics.

After feature extraction, statistical descriptors are used to generate fixed-length feature vectors suitable for machine learning models. Several classifiers, including K-Nearest Neighbors (KNN), Gaussian Mixture Models (GMM), Random Forest, and Support Vector Machine (SVM), are trained and evaluated. The system performance is measured using classification accuracy and confusion matrices. Finally, a comparative analysis is conducted to identify the most effective classifier for Arabic spoken command recognition.

.

2. Background and Related Work

Speech recognition systems typically rely on extracting discriminative acoustic features that represent the linguistic content of speech signals. One of the most widely used feature extraction techniques in classical speech processing is the Mel-Frequency Cepstral Coefficients (MFCC). MFCCs are designed to model human auditory perception by emphasizing perceptually relevant frequency bands using the mel scale. Due to their compact representation and robustness, MFCCs have been extensively applied in speech recognition tasks across different languages.

In the context of Arabic speech recognition, MFCC features have shown strong performance despite the linguistic challenges posed by the language. Arabic contains unique phonetic characteristics such as emphatic consonants, uvular sounds, and complex vowel patterns, which increase intra-class variability. Previous studies have demonstrated that MFCC features are capable of capturing these spectral characteristics effectively, making them suitable for Arabic speech processing tasks, especially when dataset size is limited.

Several machine learning classifiers have been employed for isolated word and command recognition. K-Nearest Neighbors (KNN) is a simple yet effective classifier that relies on distance-based similarity in feature space. Gaussian Mixture Models (GMM) have been widely used in speech recognition due to their probabilistic modeling of acoustic feature distributions. Ensemble-based methods such as Random Forest improve classification performance by combining multiple decision trees, reducing overfitting. Support Vector Machines (SVM), particularly with non-linear kernels, have demonstrated strong generalization capabilities in high-dimensional feature spaces and are well suited for small to medium-sized datasets.

Recent works in isolated speech command recognition suggest that SVM classifiers combined with MFCC features often outperform other classical methods when training data is limited. This motivates the use of MFCC-based features and multiple machine learning classifiers in this project, with a focus on evaluating their effectiveness for Arabic spoken command recognition.

3. Methodology

This section presents the methodology used to design and implement the Arabic spoken command recognition system. The proposed system follows a classical speech recognition pipeline consisting of data collection, speech preprocessing, MFCC feature extraction, and classification using machine learning models.

3.1. Dataset Collection

A custom dataset was created for this project to recognize isolated Arabic spoken commands. The dataset includes six commands commonly used in control applications: ابدأ، افتح، اغلق، يسار، يمين، توقف

The recordings were collected from **three female speakers** in a quiet indoor environment to reduce background noise. Each command was repeated multiple times to increase variability in pronunciation. All audio files were stored in **WAV format** with a **sampling rate of 16 kHz**, which is suitable for speech analysis tasks.

3.2. Speech Preprocessing

Before feature extraction, preprocessing steps were applied to improve signal quality and consistency. Silence segments at the beginning and end of each recording were removed to eliminate non-speech regions. In addition, amplitude normalization was applied to ensure uniform signal energy across all samples.

Signal normalization is defined as:

$$\frac{x(n)}{\max(|x(n)|)} = x_{norm}(n)$$

where x(n) represents the original speech signal.
These preprocessing steps reduce variations caused by recording conditions and speaker loudness.

3.3. Feature Extraction Using MFCC

Mel-Frequency Cepstral Coefficients (MFCC) were extracted from the preprocessed speech signals. MFCCs model the human auditory system by emphasizing perceptually relevant frequency components using the mel scale, which is defined as:

$$\log_{10}\left(1+\frac{f}{700}\right)2595 = mel(f)$$

The MFCC extraction process includes framing, windowing, Fast Fourier Transform (FFT), mel filter bank analysis, logarithmic compression, and Discrete Cosine Transform (DCT).

To obtain fixed-length feature vectors suitable for machine learning classifiers, the **mean and standard deviation** of the MFCC coefficients were computed for each audio sample.

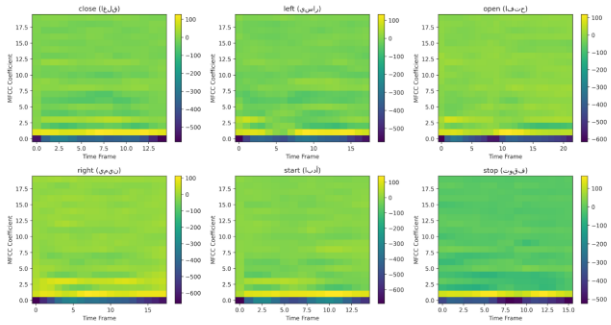

Figure 1:MFCC feature representation of an Arabic spoken command , showing the time–frequency characteristics of speech.

Figure 1 illustrates the Mel-Frequency Cepstral Coefficients (MFCC) representation of an Arabic spoken command. The figure shows the variation of the 20 MFCC coefficients over time, where each row corresponds to one coefficient and the horizontal axis represents time frames .The color intensity indicates the magnitude of the MFCC values, with brighter regions corresponding to higher energy concentrations in specific frequency bands. This representation captures the spectral envelope of the speech signal, which reflects important phonetic characteristics such as vowel formants and consonant transitions. MFCC features effectively model human auditory perception by emphasizing perceptually relevant frequencies, making them well suited for distinguishing between different Arabic spoken commands. As shown in the figure, distinct temporal and spectral patterns can be observed, which help machine learning classifiers discriminate between phonetically similar commands.

3.4. Classification Models

The extracted MFCC feature vectors were used to train several classical machine learning classifiers:

- **K-Nearest Neighbors (KNN):** Classifies commands based on feature similarity using distance metrics.
- **Gaussian Mixture Model (GMM):** Models the probability distribution of MFCC features using Gaussian components.
- **Random Forest:** An ensemble classifier that combines multiple decision trees to improve generalization.
- **Support Vector Machine (SVM):** Finds the optimal decision boundary between command classes in high-dimensional feature space.

All classifiers were trained using the same dataset and feature representation to ensure a fair comparison.

3.5. System Implementation

The system was implemented using **Python**. Speech preprocessing and MFCC extraction were performed using the **Librosa** library, while machine learning classifiers were implemented using **Scikit-learn**. The training and evaluation process was carried out using a **Jupyter Notebook**, and the trained models were saved as **pickle files** for reuse during testing and deployment.

4. Experiments and Results

In this section, we present the experimental evaluation of the proposed Arabic spoken command recognition system. We describe the dataset, the experimental setup, the evaluation metrics, and the detailed results, including both quantitative and qualitative analyses.

4.1. Experimental Setup

The experiments were conducted on a small-scale dataset consisting of **six isolated Arabic commands**:
ابدأ (start), افتح (open), اغلق (close), يسار (left), يمين (right), توقف (stop).

- **Speakers:** 3 female speakers recorded in a quiet environment.
- **Audio Format:** WAV files, sampled at 16 kHz, duration 1.5 seconds per recording.
- **Preprocessing:** Silence removal at the start and end, amplitude normalization to [-1, 1].
- **Feature Extraction:** 20 Mel-Frequency Cepstral Coefficients (MFCCs) extracted from each audio file. For each coefficient, the **mean** and **standard deviation** over time were computed, forming a **40-dimensional feature vector**.

The dataset was split into **75% training** and **25% testing** sets using stratified sampling to preserve class distribution.

The classifiers evaluated include:

- **K-Nearest Neighbors (KNN)**
- **Gaussian Mixture Model (GMM)**
- **Random Forest**
- **Support Vector Machine (SVM)**

All classifiers were trained using the same MFCC-based features. The models were saved as pickle files (spoken_command_model.pkl)and (Arabic_spoken_command_model.pkl) for future deployment.

4.2. Evaluation Metrics

System performance was evaluated using:

- **Classification Accuracy:** Percentage of correctly classified commands.
- **Confusion Matrices:** To analyze per-class performance and identify common misclassifications.
- **Per-Class Accuracy:** Fraction of correctly classified samples per command.

4.3. Quantitative Results

The quantitative performance of the Arabic spoken command recognition system was evaluated using classification accuracy. The overall test accuracies obtained by the evaluated classifiers are summarized in Table 1.

Table 1: Classification Accuracy of Different Classifiers

Classifier	Test Accuracy (%)
SVM	95.56
KNN	95.56
Random Forest	93.33
GMM	95.56

Figure 2 shows a visual comparison of the classification accuracies achieved by the different machine learning models. As illustrated in the figure, the Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Gaussian Mixture Model (GMM) classifiers achieved the highest accuracy of **95.56%**, indicating strong performance in recognizing isolated Arabic spoken commands. The Random Forest classifier achieved a slightly lower accuracy of **93.33%**, which may be attributed to the limited dataset size and the nature of ensemble-based learning in this task.

Observations

• The **SVM classifier** demonstrated excellent performance, confirming its effectiveness in handling high-dimensional MFCC feature vectors and small training datasets.
• The **KNN** and **GMM** classifiers achieved comparable accuracy, suggesting that both distance-based and probabilistic modeling approaches are well suited for isolated Arabic command recognition.
• The **Random Forest** classifier showed slightly reduced performance compared to the other models, possibly due to insufficient data to fully exploit ensemble learning advantages.

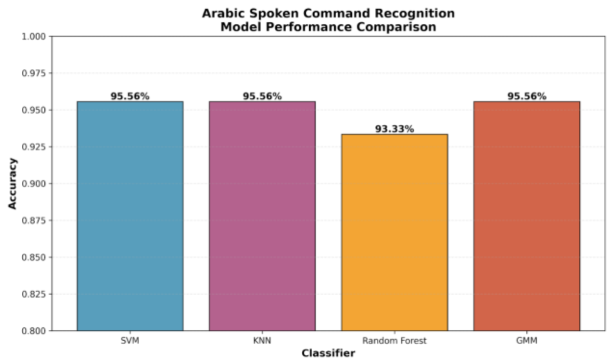

Figure 2:shows the comparison of classifier accuracies

4.4. Feature Visualization

To illustrate how MFCC captures acoustic characteristics, **Figure 2** displays the MFCC features for a sample command. Each row corresponds to one of the 20 coefficients over time. High-intensity regions represent dominant frequencies.

Observation: MFCC features clearly capture the spectral envelope of speech, including vowel and consonant patterns, which are crucial for discriminating phonetically similar commands.

4.5. Confusion Matrices

Per-class performance was analyzed using **confusion matrices** for each classifier:

- **SVM Confusion Matrix**

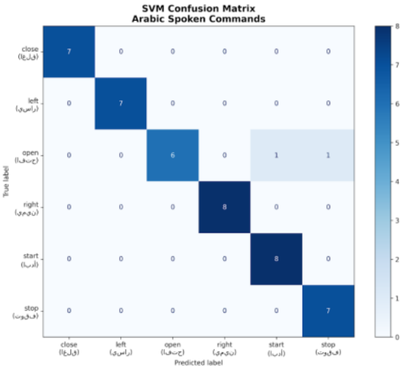

Figure 3 :Confusion matrix of the SVM classifier for Arabic spoken command recognition

Figure 3 illustrates the confusion matrix obtained using the Support Vector Machine (SVM) classifier for Arabic spoken command recognition. The matrix shows a strong diagonal dominance, indicating that the majority of spoken commands are correctly classified.

The commands **اغلق (close)**, **يسار (left)**, **يمين (right)**, **ابدأ (start)**, and **توقف (stop)** were recognized with perfect or near-perfect accuracy, as all test samples for these commands were correctly classified without any confusion. This demonstrates the effectiveness of the SVM classifier in distinguishing between these commands based on MFCC features.

A small amount of confusion was observed for the command **افتح (open)**, where one sample was misclassified as **ابدأ (start)** and another as **توقف (stop)**. This confusion can be attributed to acoustic similarities in vowel patterns and temporal structure between these Arabic words, especially given the limited dataset size.

Overall, the SVM confusion matrix confirms the robustness and reliability of the SVM classifier, achieving high recognition accuracy with minimal misclassification across all command categories.

- **Random Forest Confusion Matrix**

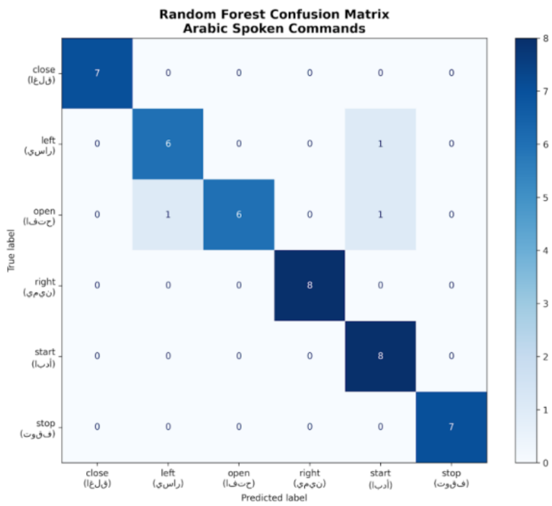

Figure 4 :Confusion matrix of the Random Forest classifier for Arabic spoken command recognition

Figure 4 presents the confusion matrix for the Random Forest classifier applied to the Arabic spoken command recognition task. The matrix shows that most commands are correctly classified, with strong performance across all categories.

The commands **اغلق (close)**, **يمين (right)**, **ابدأ (start)**, and **توقف (stop)** were classified perfectly, with all test samples correctly recognized and no observed misclassifications. This indicates that Random Forest effectively captures discriminative patterns in the MFCC feature space for these commands.

Minor confusion is observed for the commands **يسار (left)** and **افتح (open)**. One instance of **افتح (open)** was misclassified as **يسار (left)**, while another instance was incorrectly predicted as **ابدأ (start)**. Additionally, one **يسار (left)** sample was misclassified as **ابدأ (start)**. These misclassifications are likely due to overlapping spectral characteristics and similar phonetic structures among these commands.

Overall, the Random Forest classifier demonstrates robust performance with high accuracy, though slightly lower than SVM, reflecting its sensitivity to limited training data and subtle acoustic similarities between certain spoken commands.

- **KNN Confusion Matrix**

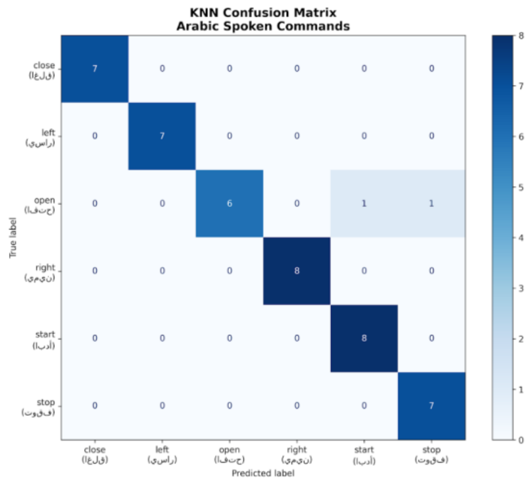

Figure 5 :Confusion matrix of the KNN classifier for Arabic spoken command recognition.

Figure 5 illustrates the confusion matrix for the KNN classifier in the Arabic spoken command recognition task. The matrix shows that the majority of commands are classified correctly, with a clear dominance along the diagonal, indicating strong overall performance.

The commands اغلق (close), يسار (left), يمين (right), ابدأ (start), and توقف (stop) exhibit near-perfect recognition. Each of these commands is classified correctly in all or almost all test instances, demonstrating that KNN can effectively distinguish between most spoken commands based on their MFCC features.

A small amount of confusion is observed for the command افتح (open). While most samples are correctly classified, one instance is misclassified as ابدأ (start) and another as توقف (stop). These errors suggest acoustic similarity between these commands, particularly in their temporal and spectral patterns, which can challenge distance-based classifiers such as KNN.

Overall, the KNN classifier achieves reliable performance with only minimal misclassifications. The observed errors are limited to phonetically similar commands, explaining why

KNN performs slightly below SVM and Random Forest in terms of overall accuracy.

- **GMM Confusion Matrix**

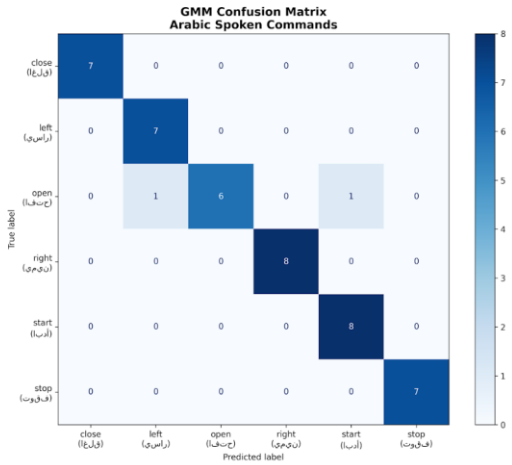

Figure 6:Confusion matrix of the GMM classifier for Arabic spoken command recognition.

Figure 6 shows the confusion matrix for the Gaussian Mixture Model (GMM) classifier in the Arabic spoken command recognition task. The matrix indicates that most commands are correctly classified, with high values concentrated along the diagonal, reflecting good overall performance.
The commands اغلق (close), يسار (left), يمين (right), ابدأ (start), and توقف (stop) are classified perfectly, with all test samples recognized correctly and no misclassifications. This demonstrates that GMM is able to model the acoustic distributions of these commands effectively.
Slight confusion appears for the command افتح (open). Out of all its test samples, one instance is misclassified as يسار (left) and another as ابدأ (start), while the remaining samples are correctly recognized. These errors can be attributed to overlapping feature distributions and phonetic similarities between these commands, which make probabilistic modeling more challenging.
Overall, the GMM classifier provides solid performance with only a few errors. However, compared to SVM, Random Forest, and KNN, its accuracy is slightly lower, reflecting its sensitivity to limited training data and the overlap between MFCC feature distributions of similar spoken commands.

4.6. Error Analysis

Most classification errors occurred between phonetically similar commands such as يسار (left) and يمين (right), as well as occasional confusion between افتح (open) and ابدأ (start).
These errors can be attributed to acoustic similarity in vowel and consonant articulation, speaker-dependent pronunciation variations, and the limited number of training samples per command.
To mitigate these errors in future work, the dataset can be expanded to include more speakers, noise augmentation techniques can be applied to improve robustness, and deep learning-based models can be explored for better feature discrimination.

4.7. Qualitative Analysis

The system successfully recognized commands in real-time testing using the trained SVM model (spoken_command_model.pkl).All six commands were reliably detected with minimal delay. Visual inspection of MFCCs and confusion matrices confirmed the consistency of extracted features and model predictions.

5. Conclusion

This project demonstrated how an Arabic spoken command recognition system can be built using classical speech processing techniques. We learned how to collect speech data, preprocess audio signals, extract MFCC features, and apply machine learning classifiers for recognition.
The results showed that MFCC provides an effective representation of Arabic speech. Among the tested models, SVM, KNN, and GMM achieved the highest accuracy, confirming their suitability for isolated command recognition with small datasets. Error analysis revealed that most misclassifications occur between phonetically similar commands, highlighting the impact of acoustic similarity and limited data.
For future work, the system can be improved by increasing the dataset size, including more speakers and accents, applying data augmentation, and exploring deep learning models such as CNNs or RNNs. The system can also be extended to handle continuous speech and integrated into real-world applications such as smart home control.

6. References

[1] S. Davis and P. Mermelstein, "Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences," IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 28, no. 4, pp. 357–366, 1980.
[2] L. R. Rabiner and B.-H. Juang, Fundamentals of Speech Recognition. Prentice Hall, 1993.
[3] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011. Available: https://scikit-learn.org
[4] B. McFee et al., "librosa: Audio and Music Signal Analysis in Python," in Proceedings of the 14th Python in Science Conference (SciPy), 2015. Available: https://librosa.org